

DSA Day 11: Polynomials with Linked Lists

Book: Data Structures Through C in Depth - Ch3 §3.9-3.10 · pp 98-107 | Code in Python

Overview

Represent a polynomial as a list of terms (coeff, exp). Adding merges terms by exponent.

Key concepts to master

- term(coeff,exp,next)
- Merge by descending exponent
- Array-list vs linked-list trade-offs

Python implementation

```
def add_poly(p,q):
    res=[]
    i=j=0
    while i<len(p) and j<len(q):
        if p[i][1]==q[j][1]:
            res.append((p[i][0]+q[j][0],p[i][1])); i+=1; j+=1
        elif p[i][1]>q[j][1]: res.append(p[i]); i+=1
        else: res.append(q[j]); j+=1
    return res+p[i:]+q[j:]
```

Complexity

Addition $O(n+m)$.

Common pitfalls

- Dropping terms whose coeff becomes 0
- Unsorted exponents

Practice

- Add $3x^2+2x+5$ and $4x^2+x$
- List 2 pros & 2 cons of linked lists

